

How Journalists Make Maps

Lilou LAGREDE DRAI

Polytechnique Montréal
lilou.lagrede--drai@polymtl.ca

Christophe HURTER

ENAC
Christophe.hurter@enac.fr

Thomas HURTUT

Polytechnique Montréal
thomas.hurtut@polymtl.ca

Introduction & Abstract

Contextual Constraints

- Tight deadlines
- Mobile-first display [2]
- Performance + accessibility

Need for Study

- How maps are actually used?
- What limits their design?
- How to support journalists?

Maps in Journalism

Communicative Power

- Transform data → spatial narrative
- Reveal patterns, disparities
- Foster immersive storytelling

Production Challenges

- Data prep = 80% of time
- Requires technical + editorial effort
- Limited automation

While cartography and journalism share goals of clear communication and reader engagement, their integration in media production remains challenging. Journalists struggle to create geographic visualizations aligned with their narratives, despite rising expectations for cartographic fluency. The rise of no-code tools like Datawrapper has transformed this landscape, enabling faster production and prompting shifts in editorial workflows. Based on eight interviews and a corpus of 3,500 maps, we examine how cartographic integration reshapes journalistic practice. Maps fulfill various functions—locating events, revealing patterns, and supporting arguments through spatial data. This study addresses how maps are produced in newsrooms, what constraints shape their use, and how collaborative processes impact publication. Our findings aim to support tool development that enhances accessibility, improves production workflows, and suggests contextually appropriate map forms.

Our methodology

8 interviews with data journalists

Identify

Mapping motivations, production workflows

Understand

Interplay between article content and maps, practical design considerations

Collection of all maps produced by a national media

Analysis

3,500 maps created by journalists at Radio-Canada between 2018 and 2025

Observation

Examination of production trends, publication patterns, and visual design choices.

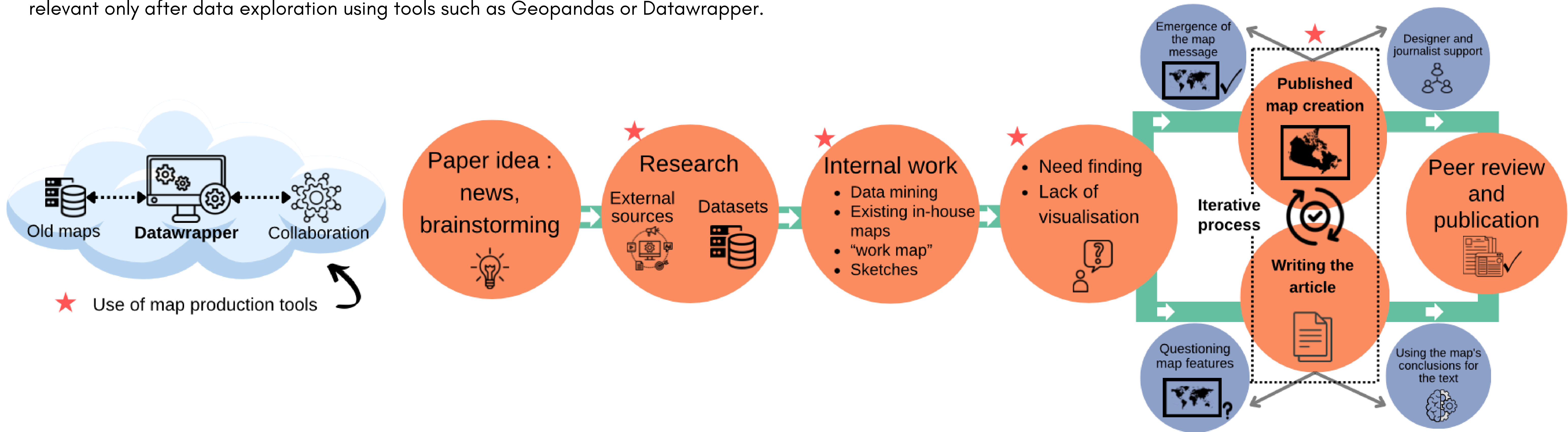
When and Why Maps Are Used

- Maps act as essential non-textual aids to understanding
- Describing trajectories, specific locations, or comparisons between geographically affected areas

Research question : How do differences between short- and long-term projects, varying levels of cartographic expertise among journalists, and the demanding constraints of the journalistic field influence the way maps are produced and used in newsrooms?

Map-enhanced article production : The Pipeline

The collaborative production process reveals that while journalists initiate map integration to convey their intended message, they don't always execute the technical production personally, particularly due to experience limitations. Journalists highlight that cartographic intent varies widely. For some, mapping precedes data analysis (topics like elections, weather events, or conflicts) where spatial representation is both central and popular with readers. For others, geographic visualization becomes relevant only after data exploration using tools such as Geopandas or Datawrapper.



Publication Patterns

Analysis of this map database reveals a key distinction: **not all produced maps are published**. This gap highlights the dual role of mapping in journalism :

- an internal analytical tool
- a means of public communication

Follow-up interviews confirmed that journalists often create maps for themselves, for data exploration, using them to extract insights that inform the article, even if the visual itself is not intended to be published.

This disparity highlights a strong editorial preference for maps that support spatial orientation over those that present more complex data relationships.

Location maps (62.2%) Aim to help readers identify a specific place—be it unfamiliar or familiar—by visually encoding geographical points or areas	Choropleth maps (30.1%) Display the distribution of aggregated values across regions, such as election results.	Symbol maps (7.6%) Using variations in size, color, or shape, highlight specific points of interest.

Discussion & Conclusion

Shift in media cartography

Toward more **collaborative workflows**, supported by platforms like Datawrapper and shared map libraries

Persistent constraints

Deadlines, limited resources, and insufficient technical tools **influence production decisions** and lead to implicit standardization

Improved integration and accessibility

A need for **clearer alignment between editorial intent and cartographic form** : recommendation map tool or accessibility assistant

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